

Name: _____

Class: _____

Date: _____

Lab 10: Loop-de-Loop

Python Lab · Section 3.8 Iteration



Goal

Repeat code with for and while loops, use accumulators, and avoid infinite loops.

Warm-up: Predict

```
total = 0
for i in range(1, 5):
    total = total + i
print(total)
```

Predict the output. Which numbers does range(1, 5) include?

AP Connection

- for i in range(n) ≈ REPEAT n TIMES.
- while condition ≈ REPEAT UNTIL (NOT condition) — flip it!
- for item in list ≈ FOR EACH item IN list.

Build It

Step 1. Countdown with a while loop.

```
n = 10
while n > 0:
    print(n)
    n = n - 1
print("Liftoff!")
```

Step 2. Accumulator: add up the numbers 1 to 100.

```
total = 0
for i in range(1, 101):
    total = total + i
print(total)
```

Step 3. Guessing game: keep asking until the user guesses the secret number.

```
secret = 7
guess = int(input("Guess: "))
while guess != secret:
    guess = int(input("Try again: "))
print("You got it!")
```

Step 4. Make an infinite loop on purpose (then stop it!). What made it infinite?

```
x = 1
while x != 10:
    x = x + 2
```

Date:

Build It (continued) · Level Up

How do you change a Python while condition into an AP REPEAT UNTIL condition?